



BAULKHAM HILLS HIGH SCHOOL

YEARLY 2013

YEAR 11 TASK 4

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 90 minutes
- Write using black or blue pen
- Board-approved calculators may be used
- Show all necessary working in Questions 11-13
- Marks may be deducted for careless or badly arranged work

Total marks – 55

Exam consists of 8 pages.

This paper consists of TWO sections.

Section 1 – Pages 2-4 (10 marks)

Questions 1-10

- Attempt Questions 1-10
- Allow about 15 minutes for this section

Section II – Pages 5-8 (45 marks)

- Attempt questions 11-14
- Allow about 1 hour and 15 minutes for this section

Section I

10 marks

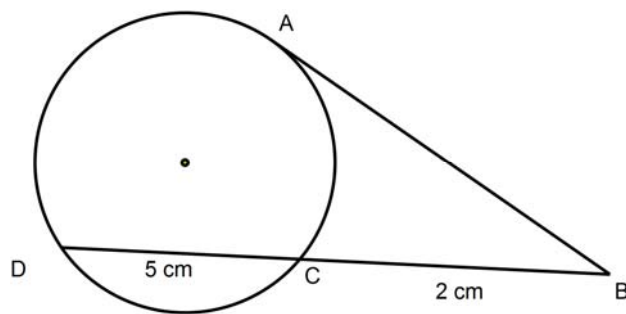
Attempt questions 1-10

Allow about 15 minutes for this section

Use the multiple choice answer sheet for questions 1-10

1. Make G the subject of the formula $E = 1 + \sqrt{\frac{G}{R}}$
(A) $G = R(1 + E)^2$ (B) $G = R(1 + E^2)$ (C) $G = R(1 - E^2)$ (D) $G = R(1 - E)^2$
2. The number of different arrangements of the letters of the word **SERVICES** which begin and end with the letter **S** are
(A) $\frac{6!}{(2!)^2}$ (B) $\frac{8!}{(2!)^2}$ (C) $\frac{6!}{2!}$ (D) $\frac{8!}{2!}$

3.



AB is a tangent to the circle, $BC = 2\text{cm}$ and $CD = 5\text{cm}$. The length of AB is

- (A) $\sqrt{10}$ (B) $\sqrt{14}$ (C) $\sqrt{20}$ (D) $\sqrt{35}$
4. If $t = \tan \frac{\theta}{2}$ then $4 \sin \theta - 3 \cos \theta = 4$ results in the equation
(A) $7t^2 - 8t + 7 = 0$
(B) $4t^2 + 3t = 0$
(C) $3t^2 - 8t + 7 = 0$
(D) $t^2 - 8t + 7 = 0$

5. The number of solutions of the equation $\sin 2x = \cos x$ where $0^\circ \leq x \leq 360^\circ$ is

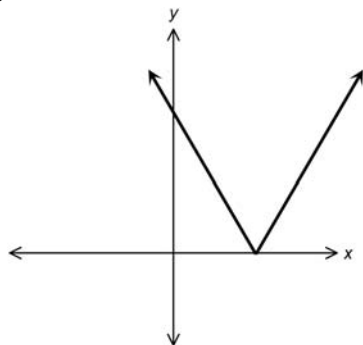
(A) 2 (B) 3 (C) 4 (D) 5

6. What is the size of the acute angle between the lines $x = 1$ and $y = 2x - 1$?

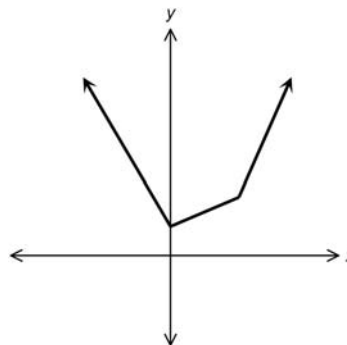
(A) $22\frac{1}{2}^\circ$ (B) 27° (C) 45° (D) 63°

7. The graph of $y = |x| + |x - 1|$ is

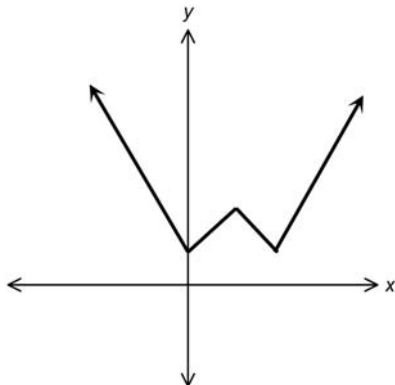
(A)



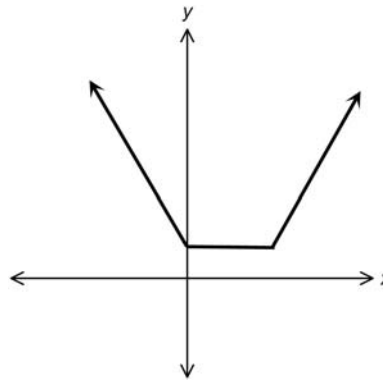
(B)



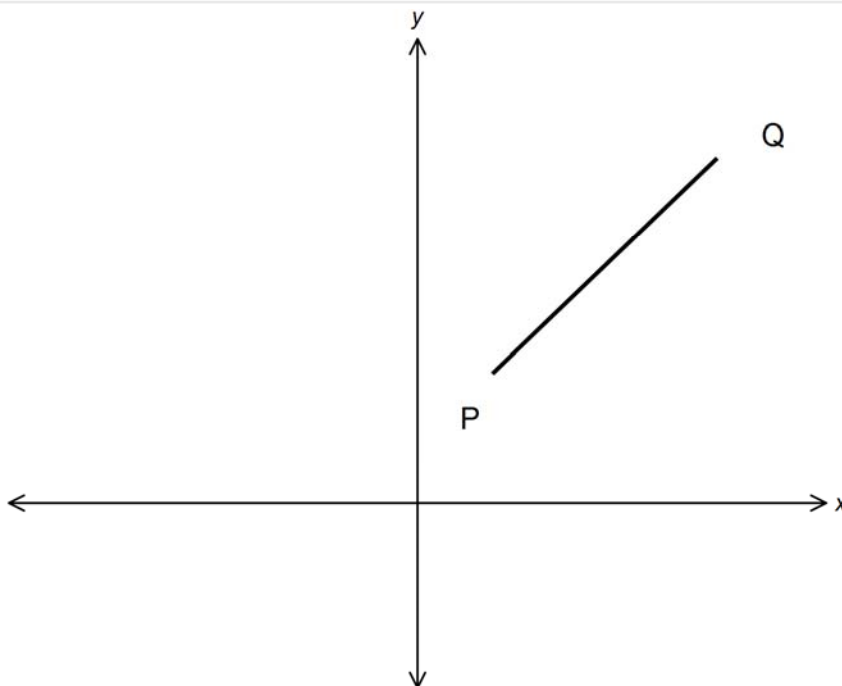
(C)



(D)



8.



T divides PQ internally in the ratio 2:1

R divides PQ externally in the ratio 2:1.

In what ratio does Q divide RT ?

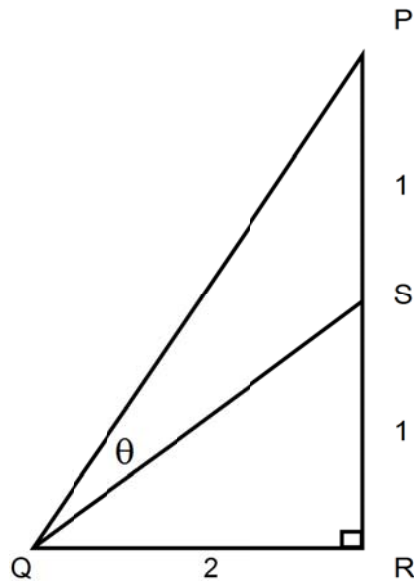
(A) 3:1

(B) 1:3

(C) 3:4

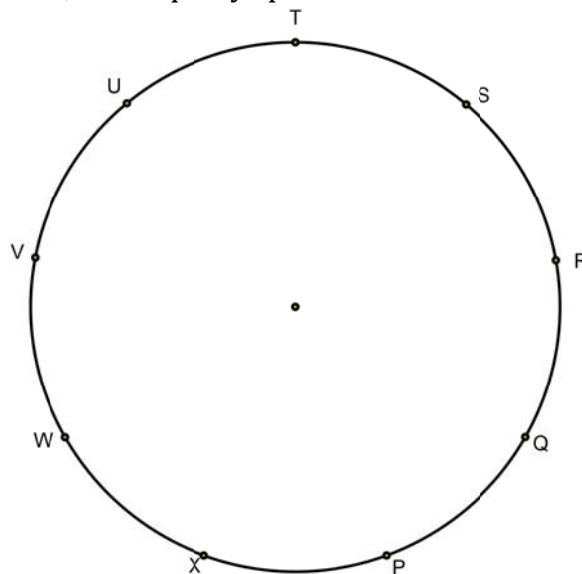
(D) 4:3

9. In triangle PQR , PS and SR both have a length of 1cm. QR is 2cm.



The value of $\tan \theta$ is

- (A) $\frac{1}{3}$ (B) $\frac{1}{\sqrt{5}}$ (C) $\tan\left(22\frac{1}{2}^\circ\right)$ (D) $\frac{1}{2}$
10. The nine points $P, Q, R, S \dots W, X$ lie equally spaced around the circumference of a circle.



The number of distinct triangles that can be formed so that the centre of the circle lies in the interior of each triangle is

- (A) 28 (B) 30 (C) 84 (D) 90

End of Section 1

Section II

45 marks

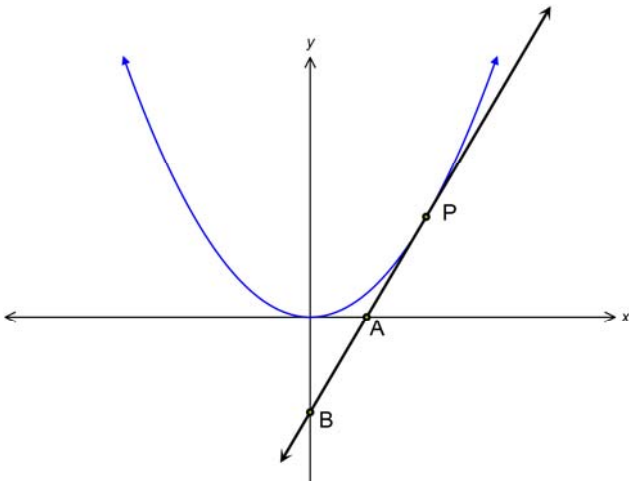
Attempt questions 11-13

Answer each question on the appropriate page of your writing booklet. Extra writing paper is available. Each piece of additional writing paper must show your number.

Clearly indicate the question number.

In Questions 11-13, your responses should include relevant mathematical reasoning and/or calculations.

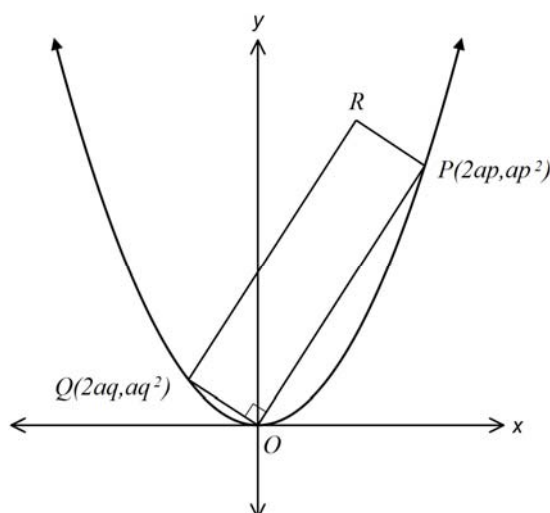
Question 11 Start on the appropriate page in your writing booklet.		Marks
a)	Find the exact value of $\sin 15^\circ$	2
b)	Find the coordinates of the point which divides the interval AB with $A(1,4)$ and $B(5,2)$ externally in the ratio 1:3.	2
c)	Find the Cartesian equation for the parametric equations $x = \cos 2t$ and $y = \cos t$	2
d)	Solve for x : $ x^2 - 5 = 5x + 9$	3
e)	Solve for x : $\frac{1}{x} < \frac{1}{x+1}$	3
f)	Find the acute angle between the tangents to $y = x^2$ at $x = -1$ and $x = 2$. Give your answer to the nearest degree.	3
End of Question 11		

Question 12 Start on the appropriate page in your writing booklet.		Marks
a)	(i) Express $\sqrt{3} \cos x - \sin x$ in the form $A \cos(x + \alpha)$ where $A > 0$ and α is acute.	2
	(ii) Hence, solve $\sqrt{3} \cos x - \sin x = -2$ for $0^\circ \leq x \leq 360^\circ$.	1
b)	Solve $\cos 2\theta + \sin^2 \theta = \frac{1}{2}$ for $0^\circ \leq \theta \leq 360^\circ$	3
c)	A group of tourists waiting to board a flight is comprised of 3 Americans, 4 Germans and 6 New Zealanders.	
	(i) How many ways can they all line up?	1
	(ii) How many ways can the Americans and the Germans line up to board the plane if no New Zealanders board?	1
	(iii) How many ways can the group line up to board the plane if the New Zealanders are included but no two New Zealanders stand next to each other?	2
d)	 P is the point $(2ap, ap^2)$ on the parabola $x^2 = 4ay$ and l is the equation of the tangent at P. The tangent at P intersects the x axis at A and the y axis at B.	
	(i) Prove that the equation of l is $y = px - ap^2$	1
	(ii) Find the coordinates of A and B .	2
	(iii) In what ratio does P divide AB ?	2
End of Question 12		

Question 13 Start on the appropriate page in your writing booklet.

Marks

a)



$P(2ap, ap^2)$ and $Q(2aq, aq^2)$ are two points on the parabola $x^2 = 4ay$. The chord PQ subtends a right angle at the origin, O . R is the fourth vertex of the rectangle $POQR$.

(i) Prove $pq = -4$

2

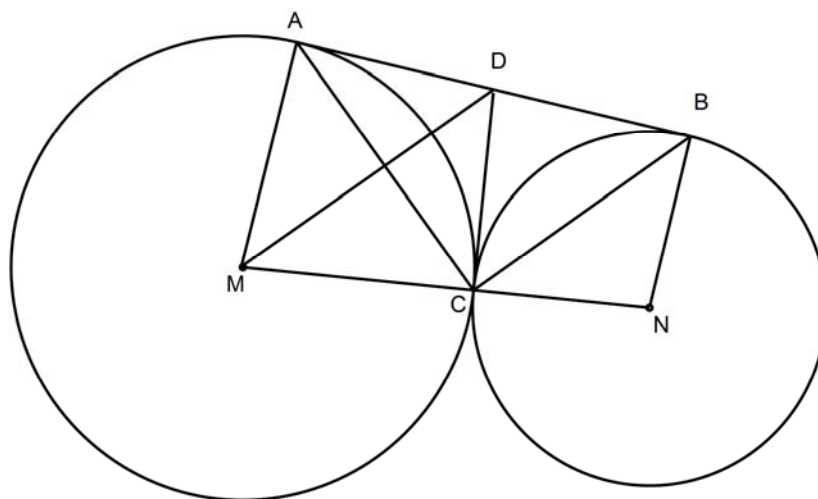
(ii) Prove that R has coordinates: $x = 2a(p + q)$, $y = a(p^2 + q^2)$

1

(iii) Prove that the Cartesian equation of the locus of R is $x^2 = 4a(y - 8a)$

2

b) In the diagram, MCN is a straight line. Circles are drawn with centre M , radius MC , and centre N , radius NC . AB is a common tangent to both circles, with the points of contact at A and B respectively. CD is a common tangent and meets AB at D .



(i) Copy the diagram

(ii) It can be proven that $\triangle ADC \parallel \triangle BNC$. (Do NOT prove this).

2

Explain why $AMCD$ is a cyclic quadrilateral.

(iii) Prove that $MD \parallel CB$

2

Question 13 continues on the following page

Question 13 (continued)

c) A bag contains 3 red balls, 2 green balls, 2 yellow balls, 2 white balls and 1 black ball. Four balls are chosen from the bag.

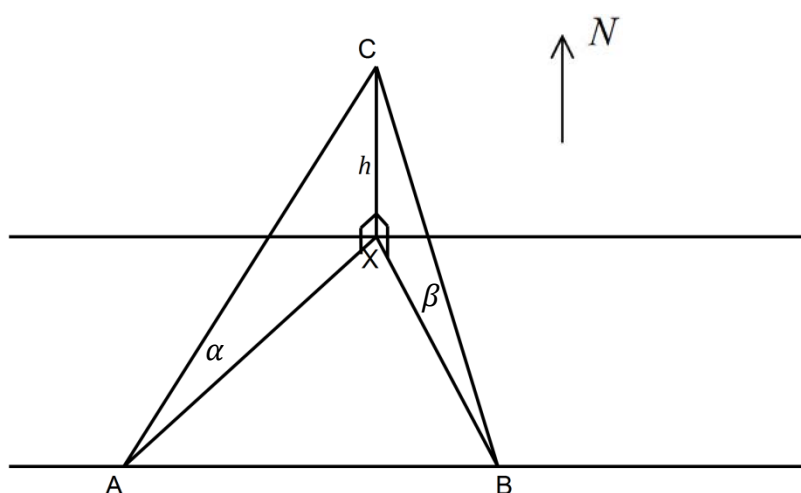
(i) How many ways are possible in which 3 of the balls are the same colour?

1

(ii) How many different combinations are possible?

2

d) A vertical mast stands on the north bank of a river with straight parallel banks running east to west.



The angle of elevation of the top of the mast is α when measured from point A on the south bank at a distance of 5 m to the west of the mast and β when measured from point B on the south bank at a distance of 3 m to the east of the mast.

Show that the height, h , of the mast is $h = \frac{4}{\sqrt{\cot^2 \alpha - \cot^2 \beta}}$

3

End of paper

1. $E-1 = \sqrt{\frac{C}{R}}$

$(E-1)^2 = \frac{C}{R}$

$C = R(E-1)^2$

$C = R(1-E)^2$

$\therefore D$

2. $\frac{S}{E} \frac{R}{V} \frac{1}{C} \frac{E}{S}$

No. of ways = $\frac{6!}{2!}$

$\therefore C$

3. $AB^2 = BC \cdot BD$

$= 2 \times 7$

$= 14$

$AB = \sqrt{14}$

$\therefore B$

4. $\frac{4 \times 2t}{1+t^2} - 3 \cdot \frac{(1-t)^2}{1+t^2} = 4$

$\frac{8t - 3 + 3t^2}{1+t^2} = 4$

$8t - 3 + 3t^2 = 4 + 4t^2$

$0 = t^2 - 8t + 7$

$\therefore D$

5. $2 \sin \lambda \cos \lambda = \cos \lambda$

$2 \sin \lambda \cos \lambda - \cos \lambda = 0$

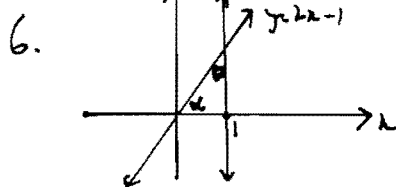
$\cos \lambda (2 \sin \lambda - 1) = 0$

$\cos \lambda = 0 \quad \sin \lambda = \frac{1}{2}$

$\lambda = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{\pi}{6}, \frac{5\pi}{6}$

$\therefore 4$ solutions

$\therefore C$



$\tan \alpha = 2$

$\alpha = 63^\circ$

$\theta = 90^\circ - 63^\circ$

$\theta = 27^\circ$

$\therefore A$

7. Between $x=0$ and 1 or $0 \leq x < 1$

$y = x + -(x-1)$

$y = 1$

$\therefore D$

8.

$RP:RQ$

$2:1$

$\therefore PQ=QR$

$\therefore Q$ divides PR in ratio $3:1$

$\therefore A$

9.

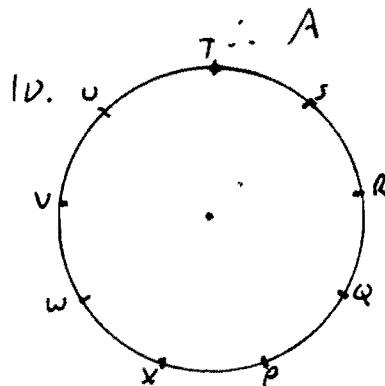
$\tan(\theta + \alpha) = \frac{2}{1}$

$\frac{\tan \theta + \tan \alpha}{1 - \tan \theta \tan \alpha} = 1$

$\tan \theta + \frac{1}{2} = 1 - \tan \theta \cdot \frac{1}{2}$

$\frac{3}{2} \tan \theta = \frac{1}{2}$

$\tan \theta = \frac{1}{3}$



Total possible $\Delta = {}^9C_3 = 84$

Determine triangles not containing centre.

From T, choose T and 2 others from U, V, W, X ie ${}^4C_2 = 6$

or choose T and 2 others from S, R, P, Q ie ${}^4C_2 = 6$

or choose T and 2 others from V, U, S, R ie ${}^4C_2 = 6$

But there are 9 vertices $\therefore (6+6+6) \times 9 = 162$

But T (and SR), S (and TR), R (and TS) are all the same $\therefore \frac{162}{3} = 54$.

$84 - 54 = 30 \therefore B$

11 a) $\sin 15^\circ = \sin (60^\circ - 45^\circ)$
 $= \sin 60^\circ \cos 45^\circ - \cos 60^\circ \sin 45^\circ \checkmark$
 $= \frac{\sqrt{3}}{2} \cdot \frac{1}{\sqrt{2}} - \frac{1}{2} \cdot \frac{1}{\sqrt{2}}$
 $= \frac{\sqrt{3}}{2\sqrt{2}} - \frac{1}{2\sqrt{2}}$
 $= \frac{\sqrt{3}-1}{2\sqrt{2}} \checkmark$

b) $A(1,4) \quad B(5,2) \quad \checkmark$
 $-1 : 3$
 $x = \frac{3-5}{3-1} \quad y = \frac{12-2}{3-1}$ (1) for a single error
 $x = -\frac{2}{2}, y = \frac{10}{2}$ (1) if correct
 $\therefore P$ is $(-1, 5) \checkmark$ internal coordinates

c) $x = \cos 2t$
 $y = \cos t$
 $x = 2\cos^2 t - 1 \checkmark$
 $x = 2y^2 - 1 \checkmark$ (ISE) for a (no cos)

d) $|x^2 - 5| = 5x + 9 \quad \underline{\text{NB}} \quad 5x + 9 \geq 0$
 $5x \geq -9$
 $x \geq -\frac{9}{5}$

$x^2 - 5 = 5x + 9 \quad x^2 - 5 = -5x - 9$
 $x^2 - 5x - 14 = 0 \quad x^2 + 15x + 4 = 0$
 $(x-7)(x+2) = 0 \quad (x+4)(x+1) = 0$
 $x = \cancel{7}, 7 \checkmark \quad x = \cancel{-4}, -1 \checkmark$
 $\therefore x = 7 \quad x = -1$

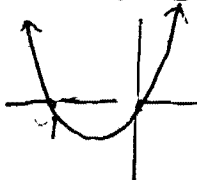
$\therefore x = -1 \text{ and } 7$

(2) if $x = 7$ only or $x = -1$ only

(1) if $x = -2, 7$ or $x = -4, -1$

e) $\frac{1}{x} < \frac{1}{x+1} \quad (x \neq 0, -1)$

$x(x+1)^2 < x^2(x+1) \checkmark$

$x(x+1)[x+1-a] < 0$
 $x(x+1) < 0 \checkmark$

 (NB squaring both sides max 1 mark)
 $\therefore -1 < x < 0 \checkmark$

f) $y = x^2$
 $\frac{dy}{dx} = 2x$
 (1) correct gradient

At $x = -1, m_1 = -2$

At $x = 2, m_2 = 4 \checkmark$

$\tan \theta = \left| \frac{4 - (-2)}{1 + 4(-2)} \right|$ (1) correctly substitutes
 $= \left| \frac{6}{-7} \right|$

$= 40^\circ 36' 4.6... \checkmark$

$\approx 41^\circ$ nearest degree

12 a(i) $A = \sqrt{(\sqrt{3})^2 + 1^2}$
 $= 2 \checkmark$

$\sqrt{3} \cos x - \sin x \equiv 2 \cos(x + \alpha)$

$\frac{\sqrt{3}}{2} \cos x - \frac{1}{2} \sin x = \cos x \cos \alpha - \sin x \sin \alpha$

$\therefore \cos \alpha = \frac{\sqrt{3}}{2}$

$\sin \alpha = \frac{1}{2}$

$\tan \alpha = \frac{1}{\sqrt{3}}$

$\alpha = 30^\circ \checkmark$

$\therefore \sqrt{3} \cos x - \sin x = 2 \cos(x + 30^\circ)$

12 a ii) $2\cos(x+30^\circ) = -2$
 $\cos(x+30^\circ) = -1$ $0 \leq x \leq 360^\circ$
 $x+30^\circ = 180^\circ$ $30^\circ \leq x+30^\circ \leq 390^\circ$
 $x = 150^\circ$ ✓

b) $\cos 2\theta + \sin^2 \theta = \frac{1}{2}$
 $\cos^2 \theta - \sin^2 \theta + \sin^2 \theta = \frac{1}{2}$ ✓ (double angle for $\cos 2\theta$)
 $\cos^2 \theta = \frac{1}{2}$
 $\cos \theta = \pm \frac{1}{\sqrt{2}}$
 $\theta = 45^\circ, 135^\circ, 225^\circ, 315^\circ$ ✓

c) i) 13! ✓

ii) 7! (= 5040) ✓

iii) Americans & Germans 7! ways
 — A — G — A — G — A — G — A —
 insert 6 New Zealanders in 8P_6 ways. ✓
 $\therefore$ Total ways = $7! \times {}^8P_6$
 $= 5040 \times 20160$
 $= 101606400$ ways ✓

12 d i) $y = \frac{x^2}{4a}$
 $\frac{dy}{dx} = \frac{2x}{4a} = \frac{x}{2a}$

At $x = 2ap$, $\frac{dy}{dx} = \frac{2ap}{2a} = p$ ✓

$y - y_1 = m(x - x_1)$
 $y - ap^2 = p(x - 2ap)$
 $y - ap^2 = px - 2ap^2$
 $y = px - ap^2$ ✓

d ii) for A: $0 = px - ap^2$
 $px = ap^2$
 $x = ap \therefore A(ap, 0)$ $p \neq 0$ ✓
 for B: $y = -ap^2$
 $\therefore B(0, -ap^2)$ ✓

d iii) let ratio be $k:1$ $A(ap, 0)$ $B(0, -ap^2)$
 $\therefore 2ap = \frac{ap + 0}{k+1}$

$k+1 = \frac{ap}{2ap}$ ✓

$k = -\frac{1}{2}$

$\therefore P$ divides AB externally in ratio $1:2$ ✓

or
 P divides AB in ratio $-1:2$.

13 a) i) $m_{AP} = \frac{ap^2}{2ap}$ $m_{AQ} = \frac{aq^2}{2aq}$
 $= \frac{p}{2}$ ✓ $= \frac{q}{2}$

Since $OP \perp OQ$ $\frac{p}{2} \cdot \frac{q}{2} = -1$

(O finds one of the gradients.)

$pq = -4$ ✓

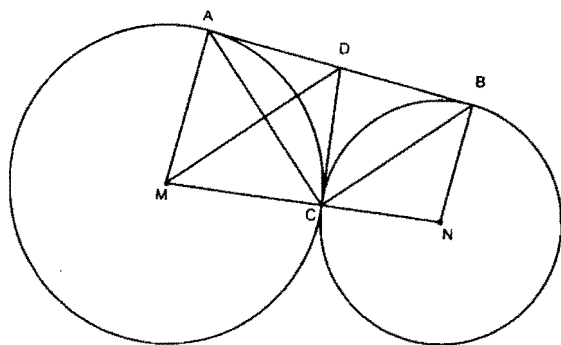
a ii) Since $POQR$ is a rectangle and opposite sides are equal and parallel }
 then R is $2aq$ units left of P and ap^2 units above P ✓

ie $x = 2ap + 2aq$ $y = ap^2 + aq^2$
 $x = 2a(p+q)$ $y = a(p^2+q^2)$

OR midpoint $R = \frac{2ap+2aq}{2}, \frac{ap^2+aq^2}{2}$ ✓
 $\therefore R$ is $\frac{2ap+2aq}{2} = 0+x_R$ $\frac{ap^2+aq^2}{2} = 0+y_R$
 $\therefore R = 2a(p+q), a(p^2+q^2)$

13a iii) $\lambda = 2a(p+q)$ $y = a(p^2+q^2)$
 $\lambda^2 = 4a^2(p^2+q^2+2pq)$ ✓
 $\lambda^2 = 4a^2\left(\frac{y}{a} + 2x - 4\right)$ ✓
 $\lambda^2 = 4a^2\left(\frac{y}{a} - 8\right)$ ✓
 $\lambda^2 = 4a(y - 8a)$ ✓
 ① attempts to use $(p+q)^2 = p^2+q^2+2pq$

13b i)



13b ii) $AM \perp AD$ (radius $\perp$ tangent)
 $\therefore \angle MAD = 90^\circ$
 Also $CM \perp CD$ (radius $\perp$ tangent)
 $\therefore \angle MCD = 90^\circ$
 In $AMCD$, $\angle MAD + \angle MCD = 90^\circ + 90^\circ = 180^\circ$
 $\therefore AMCD$ is a cyclic quadrilateral
 (opposite interior $\angle$ s are supplementary)

iii) Similarly $BNCD$ is a cyclic quadrilateral.
 ✓ $\angle DAC = \angle NCB$ (matching $\angle$ s in similar Δ s $\triangle DAC$ and $\triangle NCB$)
 ✓ $\angle DMC = \angle DAC$ ($\angle$ s at circumference standing on same chord of cyclic quad. $ADCM$)

$\therefore \angle DMC = \angle NCB$
 $\therefore MD \parallel CB$ (corresponding angles are equal)

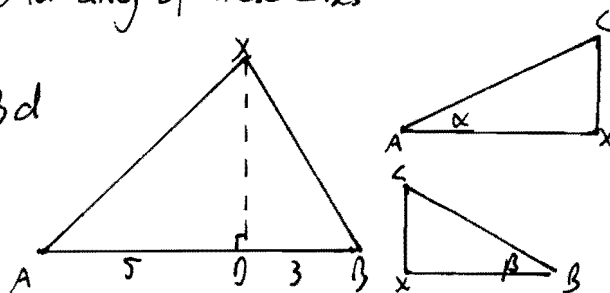
All 3 correct 2 marks
 some progress (relevant) 1 mark

13c i) All 3 red and 1 from G, Y, W, B
 i.e. ${}^4C_1 = 4$ ways. ✓

ii) 4 cases to consider
 3 red 1 other = 4 ways
 $\sqrt{2 \text{ pairs}} = {}^4C_2 = 6$ ways
 $\sqrt{1 \text{ pair, 2 different}} = {}^4C_1 \times {}^4C_2 = 4 \times 6 = 24$ ways
 $\sqrt{\text{all different}} = {}^5C_4 = 5$ ways
 Total = 39 ways ✓

① for any of these cases

13d



$\tan \alpha = \frac{CX}{AX} = \frac{h}{AX}$ $\tan \beta = \frac{CX}{BX} = \frac{h}{BX}$
 $AX = h \cot \alpha$ & $BX = h \cot \beta$ ① for either
 $AX^2 = h^2 \cot^2 \alpha$ & $BX^2 = h^2 \cot^2 \beta$
 $\therefore AX^2 - 25 = BX^2 - 9$
 $h^2 \cot^2 \alpha - h^2 \cot^2 \beta = 16$ ① for equating the expressions
 $h^2 (\cot^2 \alpha - \cot^2 \beta) = 16$
 $h^2 = \frac{16}{\cot^2 \alpha - \cot^2 \beta}$
 $\therefore h = \frac{4}{\sqrt{\cot^2 \alpha - \cot^2 \beta}}$ ① final solution